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Pressure regulation and operational control for flame display configuration

Abstract

Methods and systems for a fire device, comprising a common fuel line coupled to a fuel source via a first pressure regulator, wherein the first pressure regulator is configured to provide gaseous fuel from the fuel source to the common fuel line at a first pressure. A first fire display device is coupled to the common fuel line via a second pressure regulator, wherein the second pressure regulator is positioned downstream of the first pressure regulator and provides the gaseous fuel to the first fire display device at a second pressure that is lower than the first pressure.

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Background/Summary

FIELD

(1) The present description relates generally to methods and systems for controlling a fire display.

BACKGROUND/SUMMARY

(2) A fire pit is a vessel for containing a flame, typically used outdoors, and may be considered a type of fire display device. A torch may also be considered a type of fire display device. In some cases, it may be desirable to equip a space with multiple fire display devices including multiple fire pits and/or torches. However, pressure regulation may be challenging, and it is additionally inconvenient to procure separate fuel sources for each fire display device. Further, it may be desirable to coordinate the fire display devices to a common input, such as an audio input. Coordinating fire display devices may be result in an entertaining and/or theatrical effect. For example, it may be desired to coordinate fire bursts of the fire display devices to an audio input and with each other without a lag time.

(3) In one example, the issues described above may be addressed by a fire display system comprising a fuel source, a common fuel line coupled to the fuel source via a first pressure regulator, and a first fire display device fluidically coupled to the common fuel line via a second pressure regulator. The first pressure regulator provides the gaseous fuel to the common fuel line at a first pressure, and the second pressure regulator arranged downstream of the first pressure regulator provides the gaseous fuel to the first fire display device at a second pressure that is lower than the first pressure. Further fire display devices may additionally be coupled to the common fuel line in a similar manner as the first fire display device.

(4) In this way, pressure regulation is achieved for improved fire display device operation. Further, multiple fire display devices may share a single fuel source and a single audio input communicated without lag. Further, the number of fire display devices may be increased or decreased without the need for additional tools and minimizing a risk of release of fuel gas to the atmosphere.

Additionally, the entertainment and theatrical effect of multiple fire display devices may be increased by coordinated response to an audio input and by positioning of a pilot light at a central focus point of fuel in a burner.

(5) It should be understood that the summary above is provided to introduce in simplified form a selection of concepts that are further described in the detailed description. It is not meant to identify key or essential features of the claimed subject matter, the scope of which is defined uniquely by the claims that follow the detailed description. Furthermore, the claimed subject matter is not limited to implementations that solve any disadvantages noted above or in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a fire pit system in an environment, according to the present disclosure.
- (2) FIG. 2A schematically shows a fire pit system including connections to fire display devices.
- (3) FIG. 2B shows an example of an ignition system for a fire display device of the fire pit system of FIG. 2A.
- (4) FIG. 3A shows a fuel line and check valve of a fire pit system.
- (5) FIG. 3B shows a fuel line, check valve, and pressure regulator of a fire pit system.
- (6) FIG. 4 shows a burner configuration for a fire pit of a fire pit system.
- (7) FIG. 5 shows a control panel for a fire pit of a fire pit system.
- (8) FIG. 6A and FIG. 6B show flow charts illustrating a method for operating a fire pit system.

DETAILED DESCRIPTION

(9) The following description relates to systems and methods for a fire display system which may include multiple fire display devices. A fire display system including multiple fire display devices arranged in a display is shown in an example environment in FIG. 1. Multiple fire display devices may share a single fuel source via a common fuel line as shown in the schematic of FIG. 2A. The fire display devices may receive and distribute the fuel via the ignition system shown in FIG. 2B. The multiple fire display devices may be joined to the common fuel line via check valves and pressure regulators as detailed in FIGS. 3A-3B. The fuel may be ignited at the burner of a fire pit at a burner head including a focusing feed and pilot lite as shown in FIG. 4. The individual fire display devices of the fire display may include user input panels such as the one shown in FIG. 5. Further, the fire display may be controlled by the user input panels and/or a control hub according to the method shown in FIG. 6A-B.

(10) For purposes of discussion, the below figures are described collectively. Thus, similar elements may be labeled similarly and may not be re-introduced. FIGS. 3A-3B and 4-5 are shown approximately to scale, although other relative dimensions may be used, if desired.

(11) Turning first to FIG. 1, a fire display system **101** is shown in an example environment **100**. In the example shown by FIG. 1, the environment **100** is a large warehouse space. In other examples, however, the system environment **100** may instead be an outdoor environment, such as a backyard. The system environment **100** includes fire display devices; herein a fire display device may be a device configured to receive gaseous fuel and generate and contain an open flame. For example, fire display devices may include a plurality of torches **102a**, **102b**, **102c**, **102d** (also referred to collectively as torches **102**). Although there are four torches shown in the example at FIG. 1, it is noted that additional torches or fewer torches may be included in the system without departing from the scope of the disclosure.

(12) In addition to the torches **102**, the system environment **100** includes additional fire devices including a first fire pit **104a**, a second fire pit **104b**, and a third fire pit **104c** (also referred to collectively as fire pits **104**). As with the torches **102**, there may be additional fire pits or fewer fire

pits included in the system, in at least one example. The torches **102** and the fire pits **104** together may form a fire display system **101**. Each fire pit **104** and torch **102** may include a respective fire device controller, wireless receiver, input panel, and a battery, similar to the examples discussed further below with respect to FIG. 2A. The fire device controllers and/or wireless receivers may receive signals from a hub **110**.

(13) The hub **110** is an electronic controller that includes a processor with instructions stored in non-transitory memory that, when executed, cause the hub **110** to send control signals to control one or more of the torches **102** and the fire pits **104**. For example, the control signals sent from the hub **110** may be received at fire device controllers and wireless receivers of the respective torches **102** and fire pits **104**. Each of the torches **102** and fire pits **104** additionally includes an ignitor and at least one valve positioned therein that is configured to adjust an amount of fuel provided for ignition of the respective torch or fire pit responsive to signals (e.g., electronic signals) transmitted to the respective fire device controller by the hub **110**. For example, responsive to a given fire device controller receiving control signals from the hub **110**, the fire device controller may command the torches **102** and fire pits **104** to actuate at least one of the electric valve and the ignitor of the respective torch **102** and fire pit **104**. Via such actuation, a flame size and height may be controlled for the torches **102** and fire pits **104**.

(14) The control signals are sent from the hub **110** to one or more of the torches **102** (e.g., to the respective electronic controllers of the one or more torches **102**) in response to the processor of the hub **110** receiving input signals. The control signals may further be sent from the hub **110** to one or more of the fire pits **104** (e.g., to the respective electronic controllers of the one or more fire pits **104**) responsive to such input signals. In at least one example, the processor of the hub **110** receives input signals via one or more of a wireless receiver of the hub **110**, a hardwired connection of the hub **110**, and a user interface integrated into the hub **110** itself, where the user interface includes one or more user input devices (e.g., buttons, dials, a touch screen) to receive the input signal.

(15) In at least one example, the hub **110** may be a mobile device of a user, such as a cellular telephone or a laptop of the user. In such examples, it is noted that an application of the mobile device may be used to control the torches **102** and fire pits **104**. That is, when the hub **110** is a mobile device, an application of the mobile device may provide a display via the mobile device and receive input signals via a user interface of the mobile device (e.g., buttons, a touch screen).

(16) The input signals received at the hub **110** may include a mode selection made by a user of the fire display system and received at the hub **110**. Additionally or alternatively, a mode selection may be received at the input panels of the respective torches **102** and fire pits **104**. For example, the mode selection may include selection of a traditional mode or an audio mode. In the traditional mode, the torches **102** and fire pits **104** are operated with their respective electric valves maintained at a predetermined base position. At the predetermined base position, the electric valves of the torches **102** and the fire pits **104** are at least partially open and allow fuel to flow to their respective burners. If the electric valve of any of the torches **102** and fire pits **104** being controlled in the traditional mode is not at the base position when the traditional mode is selected, then the electric valve is first adjusted to the predetermined base position and maintained in the base position for a duration of the traditional mode. Due to the maintained position of the electric valve, a steady flame size and height is maintained in the traditional mode.

(17) In the audio mode, the torches **102** and fire pits **104** are operated with their respective electric valves being varied in coordination to an audio input, such as music. Thus, responsive to receiving a user input selecting the audio torch mode and further receiving the audio input, the hub **110** may send control signals to the audio input devices of torches **102** and fire pits **104** based on the audio input.

(18) In particular, the hub **110** may send control signals to adjust respective electric valves of the torches **102** and the fire pits **104** in coordination with the audio input. It is noted that the audio input may be received at each fire display device. For example, the audio input may be received at

the hub **110** via wirelessly streaming the audio input to the hub **110** via a mobile device or other personal computing device. In such examples, a wireless receiver of the hub **110** may receive the audio input. In examples where hub **110** is an application executed at a mobile device, the audio input may be stored at and/or wireless streamed by the mobile device and accessed by the application. As another example, the audio input may be received at the hub **110** via an aux input or other wired audio input. In such examples, a mobile device or other personal computing device may provide the audio input to the hub **110** via such an aux input or other wired audio input.

(19) The electric valve may be adjusted to positions more open than the base position of the traditional mode while in the audio mode, based on the audio input. Additionally, the electric valve may be adjusted to positions that are less open than the base position of the traditional mode while in the audio mode, based on the audio input. In this way, flame bursts and decreases in flame size may be created for the fire display. Thus, in contrast to the traditional mode, the torches **102** and fire pits **104** produce flame sizes and heights that are varied throughout the audio mode in coordination with the audio input.

(20) In at least one example, a flame boost mode may further be available for torches **102** and/or fire pits **104**, in which a maximum fuel flow is provided to a burner. In some examples, the flame boost mode may be used for purposes of heating an accessory, such as a griddle or grill attachment. The flame boost mode may also be used for purposes of producing a maximum flame height and size, which may be of interest for lighting or theatrical effect, for example. In the flame boost mode, the respective electric valve of the torches **102** or fire pits **104** is actuated to a wide open position. In at least one example, adjusting the torches **102** or fire pits **104** to operate in the flame boost mode may include adjusting a mechanical valve providing fuel to the burner to a wide open position (e.g., fully opened position), in addition to the electric valve being adjusted to the wide open position.

(21) In examples where the flame boost mode is available, it is noted that the wide open position of the electric valve is more open than the base position for the traditional mode. That is, in examples where the fire devices include the flame boost mode, the flame boost mode creates a maximum flame height and size, which is larger than the flame height and size when operating in the traditional mode.

(22) In at least one example, hub **110** allows for there to be separate control of the torches **102** and the fire pits **104**. In separate control examples, it is noted that the mode selections for each of the fire pits **104** and the torches **102** may be made individually set. Thus, each of the torches **102** and each of the fire pits **104** is able to have its own mode selected and individually controlled via the hub **110** and/or via the user input panel at each of the torches **102** and fire pits **104**. In at least one example, it is noted that a mode selected at the input panel may take priority to a mode selected at the hub. For example, if the hub **110** receives a request to operate the torches **102** and fire pits **104** in an audio mode but the user input panel of one of the fire pits **104** is set to the traditional mode, then the fire pit set to the traditional mode will be operated in the traditional mode even though the hub **110** is outputting an audio mode control signal. This allows for local control at the input panel of the torches **102** and the fire pits **104** to take priority for a particular torch or fire pit.

(23) Additionally, or alternatively, the hub **110** may control the torches **102** and the fire pits **104** collectively. In collective control examples, the hub **110** may control the torches **102** and the fire pits **104** all together to be in the same mode. For example, in collective control examples, selection of the traditional mode may result in all of the torches **102** and the fire pits **104** being set to the traditional mode. Further, in the collective control examples, selection of the audio mode may result in all of the torches **102** and the fire pits **104** being set to the audio mode. As to selection of the flame boost mode, in the collective control examples, selection of the flame boost mode may result in all of the torches **102** and the fire pits **104** being controlled to have their respective electric valves in a wide open position.

(24) Further, the hub **110** may additionally or alternatively control the torches **102** and fire pits **104**

in sub-groups. In such sub-group control, sub-groups of the torches **102** and/or the fire pits **104** may be formed for control of the sub-group to be the same. For example, in sub-group control, the hub **110** may control the torches **102** together as an all torches sub-group and may control the fire pits **104** together as an all fire pits sub-group. Thus, in this example, the mode for the all torches sub-group being selected as the traditional mode would result in the torches **102** all being set to the traditional mode. Alternatively, the mode for the all torches sub-group being selected as the audio mode would result in the torches **102** all being set to the audio mode. Similarly, in this example, the mode for the all fire pits sub-group being selected as the traditional mode would result in the fire pits **104** all being set to the traditional mode. Or, alternatively, the mode for the all fire pits sub-group being selected as the audio mode would result in the fire pits **104** all being set to the audio mode.

(25) In another sub-group control example, the hub **110** may control a portion of the torches **102** as a first torch sub-group, another portion of the torches **102** as second torch sub-group, a portion of the fire pits **104** as a first fire pit sub-group, and another portion of the fire pits **104** as a second fire pit sub-group. Moreover, a sub-group may contain both torches **102** and fire pits **104**, in at least one example.

(26) It is noted that if selection of any of the traditional mode, audio mode, and flame boost mode is also determined to initiate ignition at one or more of the torches **102** and fire pits **104**, then the hub **110** may further send a control signal to activate respective ignitors of such torches and fire pits.

(27) As described above, a fire display including multiple fire display devices may require multiple fuel containers, one for each fire display device. Further, the multiple fire display devices may not be coordinated devices able to create a desired effect, especially when the intensity of flame at each fire display device may be controlled individually or as a group or sub-group. A plurality of fire display device including respective controllers and wireless receivers as shown in FIG. 2A below may allow a coordinated display between multiple display devices and a common fuel line may allow multiple devices to be fueled by a single fuel container.

(28) Turning now to FIG. 2A, a fire display system **200** is schematically shown. The fire display system **200** includes a common fuel line **202** and a fuel source **203**. The fire display system **200** may be similar to, or the same as, the fire display system **101** described above with reference to FIG. 1. Fuel source **203** may be a rigid container holding a gaseous fuel at pressures above room pressure. The gaseous fuel may be liquefied petroleum gas (LPG), also referred to as propane. In other examples, the gaseous fuel may be natural gas. Fuel source **203** may hold the gaseous fuel at a high pressure above room pressure. For example, the pressure of the gaseous fuel inside fuel source **203** may be between 100 psi and 200 psi. A first pressure regulator **206** may be coupled to fuel source **203** to release the gaseous fuel from fuel source **203** at a pressure lower than the storage pressure. For example, first pressure regulator **206** may be configured to release gaseous fuel from fuel source **203** between 20 psi and 50 psi, though other pressure ranges are possible without departing from the scope of the disclosure. In one example, first pressure regulator **206** may release gaseous fuel at 30 psi.

(29) Fuel source **203** may be fluidically coupled to a first fire display device **204a**, a second fire display device **204b**, a third fire display device **204c**, and a fourth fire display device **204d** (also referred to as fire display devices **204**) via common fuel line **202** coupled to first pressure regulator **206**. In some embodiments, the number of fire display devices may be between one and five. Common fuel line **202** may be a flexible hose formed from a non-porous material chemically compatible with gaseous fuel. For example, common fuel line **202** may be formed from nitrile rubber or other flexible materials. For example, the common fuel line **202** may be a corrugated hose. In at least one example, when laid straight, common fuel line **202** may cover a length of up to 50 feet. In other examples, the common fuel line **202** may extend a length greater than 50 feet when laid straight. In some examples, common fuel line may include one or more junction **216** such that

common fuel line **202** extends in two different directions. For example, junction **216** may be a T-junction or a Y-junction.

(30) Common fuel line **202** may also include a bleed valve **217**. Bleed valve **217** may be positioned on common fuel line **202** at a point substantially furthest away from fuel source **203**. For example, bleed valve **217** may be positioned between junction **216** and pressure junction **208d**. Bleed valve **217** may be opened to allow air trapped in common fuel line **202** to be vented while common fuel line **202** is filled with gaseous fuel.

(31) Fire display devices **204** may include fire pits and/or torches to collectively form a fire display as described above with respect to FIG. 1. Fire display devices **204a-d** may be fluidically coupled to common fuel line **202** via pressure junctions **208a**, **208b**, **208c**, and **208d** (also referred to as pressure junctions **208**) and individual fuel lines **214a**, **214b**, **214c**, and **214d** respectively (also referred to as individual fuel lines **214**). Individual fuel lines **214** may be formed of the same material as common fuel line **202** and may be between 2 feet and 20 feet and fluidically couple pressure junctions **208** to the respective fire display devices **204**, though it is noted that other ranges for length are possible. Individual fuel lines **214** may be coupled at a first end to fire display devices **204** via a quick connect. A quick connect may be a coupler configured to fluidically join two components without the use of a tool such as a wrench. A quick connect may automatically seal upon decoupling, containing fluid within the vessel or line to which it is attached. Individual fuel lines **214** may be coupled to pressure junctions **208** at a second end. In some embodiments, the pressure junction may couple directly to fire display devices **204** via a quick connect without individual fuel lines **214**. In this way, fire display devices **204** may be coupled and decoupled from common fuel line **202** without the use of tools and minimizing an amount of gaseous fuel released to atmosphere.

(32) Pressure junctions **208** may include a check valve **212**, where the check valve **212** is coupled at a first end to common fuel line **202** and where the check valve **212** is coupled at a second end opposite the first end to a second pressure regulator **210**. Check valve **212** may be configured to ensure gaseous fuel flows only from common fuel line **202** to fire display devices **204** and not in the opposite direction. Second pressure regulator **210** may be configured to deliver gaseous fuel to the fire display device between 0.5 psi and 5 psi. The psi may be selected based on the fuel requirements of the fire display device to which it is coupled. Pressure junction **208a** may include a T-junction, coupling first fire display device **204a** to common fuel line **202** such that first individual fuel line **214a** may be perpendicular to common fuel line **202**. In some embodiments, pressure junction **208b** may also include a T-junction configured to couple second fire display device **204b** such that second fuel line **214b** may be perpendicular to common fuel line **202** and individual fuel line **214a** and individual fuel line **214b** may extend from common fuel line **202** in opposite directions. In other embodiments, pressure junctions **208c** may include a straight junction configured to couple third fire display device **204c** to common fuel line **202** such that individual fuel line **214c** extends in the same direction as common fuel line **202**.

(33) Each of fire display devices **204** may be communicatively coupled to a control module **218**. Control module **218** may include an wireless receiver **220**, a controller **222**, and a battery **224**. Wireless receiver **220** may be physically communicatively coupled (e.g., via a wire) to controller **222**. Controller **222** may be physically and communicatively coupled to fire display device **204**. Battery **224** may be electrically coupled to both wireless receiver **220** and controller **222**, thus providing power to operate both. Control module **218** may be located external to or internal to a housing of fire display devices **204**.

(34) Controller **222** may also be physically and communicatively coupled to an input panel **223**. Input panel **223** may form part of the housing of fire display devices **204** and may be used to select an audio or traditional mode as described above with respect to FIG. 1. Input panel **223** may be discussed further below with respect to FIGS. 2B and 5.

(35) Wireless receiver **220** may be configured to receive wireless signals from a control hub **226**.

The hub **226** may be similar to, or the same as, the hub **110** described above with reference to FIG.

1. In one embodiment, wireless receiver **220** may also include a microphone and be capable of receiving verbal commands. Wireless receiver **220** may send corresponding signals to controller **222** in response to verbal commands.

(36) Control hub **226** may be similar to hub **110** as described above with respect to FIG. **1**. Control hub **226** may be an application executed at a mobile device such as a cellular telephone or laptop capable of sending and receiving wireless signals. Control hub **226** may wirelessly communicate a signal to wireless receiver **220** at each of fire display devices **204**. In one embodiment, the signal may be sent via a Wi-Fi network. As one example, the signal may be an audio signal. Controller **222** may receive the audio signal from wireless receiver **220** and based on instructions stored in non-transitory memory, actuate components in response to the audio signal. Actuating components may include sending control signals to open or close electric valves or turn on electric ignitors. In another embodiment, control hub **226** may receive an audio signal and responsive to the audio signal and instructions stored in non-transitory memory, communicate commands to controller **222** via wireless receiver **220** to actuate components such as electric valves.

(37) As another example, control hub **226** may wirelessly communicate instructions to controller **222** via wireless receiver **220** to change an operation mode of the fire display device (e.g., switch from audio mode to traditional mode) and/or to switch a fire display device on or off. controller **222**.

(38) Control hub **226** may wirelessly communicate instructions to one or more fire display devices **204** simultaneously. In this way, the actions of one or more fire display devices may be synchronized.

(39) Looking briefly to FIG. **2B**, FIG. **2B** shows an example diagram **250** of the ignition configuration for one or more of the fire display devices **204**. Fire display device **204** may include a fuel passage **252** fluidically coupled to a pilot light **254** and a plurality of fuel injectors **256** via a mechanical valve **258** and an electronic valve **260**.

(40) Fuel passage **252** may include an introductory portion **252a** which may be fluidically coupled to a connector **264** positioned external to a housing **262** of fire display device **204**. Connector **264** may be a quick connect coupled to individual fuel line **214** which is in turn coupled to common fuel line **202** as discussed above with respect to FIG. **2A**. In one example, connector **264** may be positioned at a bottom side of the fire display device. Gaseous fuel traveling from introductory portion **252a** towards pilot light **254** and fuel injectors **256** may reach a first junction **266a** which may split fuel passage **252** into a mechanical portion **252b** and an electric portion **252c**. Mechanical portion **252b** may further split at second junction **266b** to include a pilot portion **252d** fluidically coupled to pilot light **254**. Mechanical portion **252b** and electric portion **252c** may rejoin at third junction **266c** to become fuel injection portion **252e** which is fluidically coupled to the plurality of fuel injectors **256**.

(41) Mechanical valve **258** may be configured to control a flow of gaseous fuel entering fuel injection portion **252e** from mechanical portion **252b**. In one example, mechanical valve **258** may be configured at mechanical portion **252b** to also control the flow of gaseous fuel to pilot light **254**. Mechanical valve **258** may be physically coupled to and controlled by user input device **268** positioned outside of housing **262**. User input device **268** may be located at input panel **223** as discussed above with respect to FIG. **2A**. For example, the user input device **268** may be a dial, where turning the dial a first direction may adjust the mechanical valve **258** to a more open position and turning the dial a second direction may adjust the mechanical valve **258** to a more closed position. Thus, via the user input device **268**, a user is able to set a base amount of fuel allowed to flow through the mechanical valve **258** into fuel injection portion **252e**.

(42) Electronic valve **260** may be configured to control a flow of gaseous fuel from electric portion **252c** into fuel injection portion **252e**. Electronic valve **260** may be an actuated valve able to regulate gaseous flow over a range of flow rates. For example, electronic valve **260** may be a butterfly

valve or other adjustable electronic valve. Electronic valve **260** may be physically and communicatively coupled to control module **218**. Control module **218** may include controller **222**, battery **224**, and wireless receiver **220** as above with respect to FIG. 2A. For example, control module **218** may receive a wireless signal **272** from a control hub (e.g., control hub **226**) with instructions to control a position of electronic valve **260** to provide flame bursts synchronized with an audio input. In this way, mechanical valve **258** may control a base amount of fuel when fire display device **204** is operated in a traditional mode while electronic valve **260** may be actuated by control module **218** to provide additional fuel resulting in flame bursts coordinated by an audio signal when in audio mode.

(43) In an alternate embodiment, mechanical valve **258** may be controlled electronically by physically and communicatively coupling to control module **218**. In this way, mechanical valve **258** may be controlled by wireless signal **272** sent to control module **218** from the control hub as well as a signal generated by user input device **268**.

(44) Gaseous fuel reaching pilot light **254** may be ignited by a spark generated by ignitor **274**. Ignitor **274** may be physically and communicatively coupled to control module **218** and may be actuated in response to a command from control module **218** and/or input panel **223**. Control module **218** may receive a signal from user input device **268** and/or a wireless signal **272** commanding ignition of pilot light **254**.

(45) Once ignited, the pilot light may ignite the gaseous fuel flowing from plurality of fuel injectors **256** as controlled by both mechanical valve **258** and electronic valve **260**. Additional details regarding plurality of fuel injectors **256** and pilot light **254** may be discussed in further detail below with respect to FIG. 4.

(46) The fuel entering fire display device **204** as described above may be sourced from a common fuel line which may be intersected by a plurality of pressure junctions to deliver fuel to a plurality of fire display devices. Turning now to FIG. 3A-3B, examples of a pressure junction **302** for coupling a fire display device to a common fuel line **304** are shown. FIG. 3A shows view **300** of pressure junction **302** including a check valve **306** while FIG. 3B shows view **350** of pressure junction **302** showing both check valve **306** and a pressure regulator **352**. Common fuel line **304** and pressure junction **302** may be similar to common fuel line **202** and pressure junctions **208** as discussed above with respect to FIG. 2A.

(47) FIGS. 3A-3B show common fuel line **304** intersected by a T-junction **308**. As discussed above, T-junction **308** may be a straight junction in some examples. T-junction **308** may be formed of a rigid non-porous material compatible with gaseous fuel. For example, T-junction **308** may be formed of a metal such as copper or stainless steel or T-junction **308** may be formed of a plastic such as polyvinyl chloride (PVC). T-junction **308** may include a first opening **310a**, a second opening **310b**, and a third opening **310c**. First opening **310a** and second opening **310b** may be coupled to common fuel line **304** via gas tight couplers. For example, T-junction **308** may be coupled to common fuel line **304** via male and female thread pipe fittings. Third opening **310c** of T-junction **308** may be likewise coupled to check valve **306**. Check valve **306** may be similar to check valve **212** of FIG. 2A. Check valve **306** may be oriented so as to allow flow of gaseous fuel indicated by arrow **312**. Check valve **306** may be a pressure type of check valve formed from a metal or plastic compatible with gaseous fuels.

(48) FIG. 3B shows check valve **306** coupled to pressure regulator **352**. Check valve **306** may be coupled to pressure regulator **352** via gas tight couplers such as male and female threaded pipe fittings. Pressure regulator **352** may receive gaseous fuel flowing from check valve **306** at a high pressure (e.g., 30 psi) and output gaseous fuel at a lower pressure (e.g., 5 psi or 0.5 psi) towards the fire display device (not shown). An outlet **354** of pressure regulator **352** may be coupled to a flexible fuel line **356** which may be similar to individual fuel lines **214** as shown in FIG. 2A. The fire display device may couple to the end of flexible fuel line **356** not coupled to pressure regulator **352** by a quick connect to deliver fuel the fire display device, eventually reaching a burner such as

the burner shown in FIG. 4.

(49) FIG. 4, shows a top-angled view **400** of burner **402**, which may be incorporated into a surface of a fire display device such as fire pits **104** of FIG. 1. Burner **402** may include a pilot light **404**, gaseous fuel injectors **408**, and top plate **414**. Pilot light **404** and gaseous fuel injectors **408** may be similar to pilot light **254** and fuel injectors **256** of FIG. 2B. Burner **402** may be fluidically coupled to a pressurized tank of gaseous fuel as described above with respect to FIGS. 2A-2B.

(50) Gaseous fuel injectors **408** may be positioned and angled such that gaseous fuel is injected to a focal point **416** which may be configured at a horizontal center of burner **402** and a vertical height **420** above the edge of an outer housing of the fire display device. Gaseous fuel injectors **408** may be fluidically coupled to gaseous fuel and to each other via a fuel injector line **410** (such as fuel injection portion **252e** of FIG. 2A), and arrayed in various configurations which allow the gaseous fuel to be injected to focal point **416**. In one example, fuel injector line **410** may form an inner circle **411** and an outer circle **412** connected by spokes **418**. Spokes **418** may extend from center post **422** to inner circle **411**. Post **422** includes a hole **423** formed therein for additional gaseous fuel flow directed towards the focal point **416**. Further, spokes **418** may fluidically couple to outer circle **412** to inner circle **411**.

(51) Pilot light **404** may include an electric ignitor (e.g., ignitor **274**) which may be actuated in response to a command from a controller or user input (such as controller **222** or input panel **223** of FIG. 2A). When ignited, pilot light **404** may be configured to provide a flame which extends above gaseous fuel injectors **408** and crosses a plane parallel to top plate **414**, intersecting the focused streams of gaseous fuel at focal point **416**. In this way, a burst effect may be readily visible, for example when fuel flowing to gaseous fuel injectors **408** is increased in response to an audio input as described above with respect to FIG. 2B. In one embodiment, pilot light **404** may be located between inner circle **411** and outer circle **412**.

(52) Top plate **414** may be configured to cover a top portion of the fire display device. Fuel injector line **410**, gaseous fuel injectors **408**, and pilot light **404** may all extend through and above top plate **414**. Top plate **414** may include a plurality of openings **424** which may improve air flow.

(53) Turning now to FIG. 5, an example **500** is shown of a side view **502** of a section of a fire display device such as fire pit **104** of FIG. 1. Side view **502** may include user input device **504**. User input device **504** may be similar to user input panel **223** as discussed above with respect to FIG. 2A. User input device **504** may be coupled to an outer housing **512** of the fire display device to be readily accessed by a user. In one example, user input device **504** may be recessed into housing **512** so as to avoid accidental user inputs.

(54) User input device **504** may include an ignite button **506**, mode button **508** and dial **510**. Ignite button **506** may be coupled to a controller such as controller **222** FIGS. 2A and 2B respectively. Pressing ignite button **506** may send a signal to the controller to actuate an ignitor in a pilot light, as discussed above with respect to FIGS. 2B and 4.

(55) Dial **510** may be physically coupled to a mechanical valve (such as mechanical valve **258** as discussed above with respect to FIG. 2B). Positions of dial **510** may control an amount of fuel passing to a fuel injection line as described above with respect FIG. 2B. In an alternate embodiment where mechanical valve **258** may be controlled electronically, dial **510** may be physically and communicatively coupled to the controller. In such an example, a user input at dial **510** may instruct the controller to open or close the mechanical valve.

(56) Mode button **508** may send a signal to the controller instructing a mode of fire display device operation. Mode button **508** may be any type of toggle switch allowing a user to toggle between three positions corresponding to three modes. The three modes may be flame off, non-boosted mode, and audio input mode. Non-boosted mode and audio input mode may correspond to different operational modes as discussed above with respect to FIG. 1. Further, the operational modes may be discussed in more detail below with respect to FIG. 6A-B

(57) Turning to FIG. 6A, a flow chart illustrating a method **600** for operating a fire display system

is shown. The fire display system may be similar to, or the same as, the fire display systems described above with respect to FIGS. 1 and 2A. The fire display system includes one or more fire display devices. The one or more fire display devices may be similar to, or the same as, the fire display device **204** described above with reference to FIGS. 2A-2B, in some examples. Instructions for carrying out method **600** and the rest of the methods included herein may be executed by a controller based on instructions stored on a memory of the controller and in conjunction with signals received from sensors of the fire display system, such as the sensors described above with reference to FIG. 1. The controller may employ actuators of the system to adjust system operation, according to the methods described below.

(58) At **602**, method **600** includes coupling one or more fire display devices to a common fuel line. The one or more fire display devices may be fire pits or torches as described above with respect to FIG. 1. The common fuel line may be a pressurized fuel line configured to deliver gaseous fuel from a common fuel source, such as fuel source **203** as described above with respect to FIG. 2A. Coupling the one or more fire display devices may include attaching the fire display device to a pressure junction via an individual fuel line using a quick connect as described above with respect to FIG. 2A.

(59) The method continues from **602** to **604** where the method includes opening the common fuel source. The common fuel source may be coupled to the common fuel line via a pressure regulator such as first pressure regulator **206** as described above with respect to FIG. 2A. As such, opening the common fuel source may result in the common fuel line being filled with gas at a pressure set by a first pressure regulator configured to control gas flow out of the common fuel source and into the common fuel line. In one example, the pressure set by the first pressure regulator may be 30 psi.

(60) At **606**, method **600** includes opening a bleed valve such as bleed valve **217** described above with respect to FIG. 2A. Opening the bleed valve may be done manually by an operator of the fire display system. Opening the bleed valve may allow air trapped in the common fuel line to escape as it is being pushed out by gaseous fuel entering the common fuel line from the common fuel source. Alternatively, the bleed valve may be automatically opened via a mechanical actuator responsive to a user input to the control hub indicating an initial coupling of the common fuel source to the common fuel line.

(61) At **608**, the method includes determining whether air has been vented from the common fuel line. Determining if the air has been vented from the common fuel line may be done manually by an operator and may include identifying a smell of propane gas coming from the bleed valve, indicating that there is no air remaining in the common fuel line. Alternatively, a user may determine a threshold amount of time for the bleed valve to remain open, after which the air is considered to be vented from the common fuel line. In one or more examples, the bleed valve may be automatically opened and held opened for the threshold amount of time to purge air from the common fuel line.

(62) If it is determined that air has not been vented from the common fuel line at **608**, the method continues to **609** and the bleed valve is maintained in an open position until air is vented as determined by smell or elapsed time as describe above with respect to step **608**. If it is determined that air has been vented from the common fuel line at **608** or at **609**, the method continues from **608** to **610** where the method includes closing the bleed valve. Closing the bleed valve may include manually adjusting a position of the bleed valve to stop the flow of gaseous fuel from leaving through the bleed valve via a user input device of the bleed valve or automatically closing the bleed valve.

(63) At **612**, method **600** includes lighting the fire display devices. After the bleed line is closed at **610**, the fire display devices may be supplied with gaseous fuel from the common fuel line at a pressure determined by a second pressure regulator coupled to the fire display device as described above with respect to FIG. 2A. Igniting the fire display device may include sending a signal to a

controller communicatively coupled to an ignitor configured within a pilot light. In some examples, the signal may be sent to the controller via a user input panel as described above with respect to FIG. 5. Additionally or alternatively, the signal may be sent from a control hub to the control module as described above with respect to FIG. 2A.

(64) Method **600** continues to **614** where operation of the fire display device is controlled as described in further detail in FIG. 6B below. After operation of the fire display device, method **600** proceeds to **616** at which point the common fuel source is closed. Closing the common fuel source may include closing a valve on the common fuel tank so that gaseous fuel is no longer delivered to the common fuel line.

(65) Turning now to FIG. 6B, a flow chart of a method **650** for controlling a fire display device coupled to a common fuel line as part of method **600** is shown. Method **650** may include executable instructions included in the non-transitory memory of a control hub and/or control module. A control module may include a wireless receiver, controller, and a battery; and is capable of receiving wireless signals from the control hub as described above with respect to FIG. 2A.

(66) At **652**, method **650** determines if there is a user input. A user input may be received via a user input device as described above with respect to FIG. 2B and FIG. 5, or via a command to a control module from a control hub as described above with respect to FIG. 2A. If there is not a user input, method **650** proceeds to **654** where the method maintains the current operational mode. If a user input is received, method **650** proceeds to **656** where it is determined if the user input is for traditional mode.

(67) If the user input is for traditional mode, a valve is adjusted in the one or more fire display devices to a base set position at **658**. The adjusted valve may be a mechanical valve such as mechanical valve **258** as described above with respect to FIG. 2B. In one embodiment, the adjustment may be made via a physically coupled user input such as a dial. Additionally or alternatively the mechanical valve may be capable of being controlled by user input to the control module via the control hub and/or user input panel. Further a base set position of an electronic valve such as electronic valve **260** described above with respect to FIG. 2B may be set in response to user input for traditional mode. A base position of electronic valve **260** may be set by instructions from the control module. If adjusted by a control hub, the base set position of one or more fire display devices may be set simultaneously. At **660**, method **650** maintains the base set position before continuing to step **672**.

(68) If the user input is not for traditional mode, method **650** determines if the user input is for audio mode. If the user input is for audio mode, method **650** proceeds to **664** where it is determined if an audio input is received. If an audio input is not received, a control valve in one or more fire display devices is set to an audio base position at **670**. The control valve setting the audio base position may be the electric valve as described above with respect to FIG. 2B. The audio base position may be set via the control hub. If an audio input is received, the control valve may be continually adjusted in one or more devices based on audio input. The audio input may be received at the one or more fire display devices via a wireless signal received by a wireless receiver from the control hub. Instructions for continual adjustment of the control valve may be given by the controller receiving the audio input from the wireless receiver. In this way the flame height in response to an audio input may be adjust at a plurality of fire display devices simultaneously. After step **668** or **670**, method **650** proceeds to **672**. Method **650** also proceeds to **672** from **662** if the user input is not for audio mode.

(69) At **672**, method **650** determines if the user input is to turn the fire display device off. If the user input is not to turn the fire display device off, then method **650** returns to **652**. If the user input is to turn the fire display device off, method **650** continues to **674** and closes all fuel valves. All fuel valves may include both the mechanical valve and the electric valve. If both the mechanical valve and electrical valve are communicatively coupled to the controller, then all fuel valves may be closed at one or more fire display devices by a command from the control module. The command

may be wirelessly issued to the control module by the control hub. If the mechanical valve is not communicatively coupled to the controller than closing all fuel valves may include physically closing the mechanical valve at the user input device as well as a command from the control hub.

(70) In this way a fire display including multiple fire display devices may be simplified and an entertainment value may be increased. One or more fire display devices may share a single common fuel tank via coupling to a common fuel line. The one or more fire display devices may be coupled via a quick connect allowing for convenient, tool free coupling and decoupling of the one or more fire display devices to the common fuel line. Additionally, the one or more fire display devices may include a pilot light configured to intersect with a focal point of gaseous fuel injectors above a top surface of the fire display device, providing an easily visible and pleasing fire effect. Further, the one or more fire display devices may be coupled to a control module configured to receive wireless signals from a control hub. The signals from the control hub may be simultaneously received by a plurality of fire display devices allowing for synchronous control.

(71) The disclosure also provides support for a fire display system, comprising: a fuel source comprising gaseous fuel, a common fuel line coupled to the fuel source via a first pressure regulator providing the gaseous fuel to the common fuel line at a first pressure, and a first fire display device fluidically coupled to the common fuel line, wherein the first fire display device is coupled to the common fuel line via a second pressure regulator arranged downstream of the first pressure regulator and providing the gaseous fuel to the first fire display device at a second pressure that is lower than the first pressure. In a first example of the system, a second fire display device is fluidically coupled to the common fuel line via a third pressure regulator arranged downstream of the first pressure regulator and providing the gaseous fuel to the second fire display device at a third pressure that is lower than the first pressure. In a second example of the system, optionally including the first example, the third pressure is equal to the second pressure. In a third example of the system, optionally including one or both of the first and second examples, the third pressure is a different pressure, lower than the second pressure. In a fourth example of the system, optionally including one or more or each of the first through third examples, the system further comprises: a bleed valve coupled to the common fuel line that is configured to vent air out of the common fuel line. In a fifth example of the system, optionally including one or more or each of the first through fourth examples, the bleed valve is positioned at an opposite end of the common fuel line as the fuel source. The disclosure also provides support for a fire display system, comprising: a fuel source comprising gaseous fuel, a common fuel line coupled to the fuel source via a first pressure regulator, wherein the first pressure regulator is configured to provide the gaseous fuel to the common fuel line at a first pressure, and a plurality of fire display devices each fluidically coupled to the common fuel line via respective pressure regulators, the respective pressure regulators configured to flow the gaseous fuel from the common fuel line to the plurality of fire display devices at pressures less than the first pressure. In a first example of the system, the system further comprises: a bleed valve coupled to the common fuel line, wherein the bleed valve is downstream of the plurality of fire display devices. In a second example of the system, optionally including the first example, the bleed valve is configured to release air out of the common fuel line. In a third example of the system, optionally including one or both of the first and second examples, the plurality of fire display devices are each further fluidically coupled to the common fuel line via respective check valves. In a fourth example of the system, optionally including one or more or each of the first through third examples, the respective pressure regulators include a second pressure regulator, and wherein the second pressure regulator is positioned between the common fuel line and a first fire display device of the plurality of fire display devices. In a fifth example of the system, optionally including one or more or each of the first through fourth examples, a fuel line extends between the second pressure regulator and the first fire display device. In a sixth example of the system, optionally including one or more or each of the first through fifth examples, each of the plurality of fire display devices comprises: a burner, an electrically actuatable gas valve

configured to adjust an amount of gaseous fuel flowed to the burner for ignition, a wireless receiver, and a controller, wherein the comprises instructions stored in non-transitory memory of the controller that are executable to: adjust a position of the electrically actuatable gas valve responsive to a request received via the wireless receiver. In a seventh example of the system, optionally including one or more or each of the first through sixth examples, the request received via the wireless receiver is from a hub of the fire display system. The disclosure also provides support for a method for operating a multi flame display, comprising: providing gaseous fuel to a common fuel line at a first pressure via a first common fuel line pressure regulator, and flowing the gaseous fuel to a plurality of flame display devices, wherein the plurality of flame display devices are fluidically coupled to the common fuel line downstream via respective pressure regulators, wherein the respective pressure regulators are positioned downstream the first common fuel line pressure regulator and provide the gaseous fuel to the plurality of flame display devices from the common fuel line to the plurality of flame display devices at lower pressures than the first pressure. In a first example of the method, the method further comprises: after providing the gaseous fuel to the common fuel line at the first pressure and prior to flowing the gaseous fuel to the plurality of flame display devices, venting air trapped in the common fuel line via a bleed valve. In a second example of the method, optionally including the first example, venting the air trapped in the common fuel line includes opening the bleed valve to release the air out of the common fuel line. In a third example of the method, optionally including one or both of the first and second examples, the method further comprises: igniting the gaseous fuel flowed to the plurality of flame display devices responsive to commands received at the plurality of flame display devices from a hub that are coordinated to an audio input. In a fourth example of the method, optionally including one or more or each of the first through third examples, the commands are received wirelessly from the hub, and wherein igniting the gaseous fuel flowed to the plurality of flame display devices includes adjusting a position of an electric valve included in each of the plurality of flame display devices. In a fifth example of the method, optionally including one or more or each of the first through fourth examples, the commands received at a first flame display device differs from the commands received at a second flame display device of the plurality of flame display devices.

(72) FIGS. 1 and 3-5 show example configurations with relative positioning of the various components. If shown directly contacting each other, or directly coupled, then such elements may be referred to as directly contacting or directly coupled, respectively, at least in one example. Similarly, elements shown contiguous or adjacent to one another may be contiguous or adjacent to each other, respectively, at least in one example. As an example, components laying in face-sharing contact with each other may be referred to as in face-sharing contact. As another example, elements positioned apart from each other with only a space there-between and no other components may be referred to as such, in at least one example. As yet another example, elements shown above/below one another, at opposite sides to one another, or to the left/right of one another may be referred to as such, relative to one another. Further, as shown in the figures, a topmost element or point of element may be referred to as a “top” of the component and a bottommost element or point of the element may be referred to as a “bottom” of the component, in at least one example. As used herein, top/bottom, upper/lower, above/below, may be relative to a vertical axis of the figures and used to describe positioning of elements of the figures relative to one another. As such, elements shown above other elements are positioned vertically above the other elements, in one example. As yet another example, shapes of the elements depicted within the figures may be referred to as having those shapes (e.g., such as being circular, straight, planar, curved, rounded, chamfered, angled, or the like). Further, elements shown intersecting one another may be referred to as intersecting elements or intersecting one another, in at least one example. Further still, an element shown within another element or shown outside of another element may be referred as such, in one example. Moreover, it is noted that reference to upstream and downstream may be made based on a direction of fuel flow from the fuel source.

(73) Note that the example control and estimation routines included herein can be used with various system configurations. The control methods and routines disclosed herein may be stored as executable instructions in non-transitory memory and may be carried out by the control system including the controller in combination with the various sensors, actuators, and other hardware. The specific routines described herein may represent one or more of any number of processing strategies such as event-driven, interrupt-driven, multi-tasking, multi-threading, and the like. As such, various actions, operations, and/or functions illustrated may be performed in the sequence illustrated, in parallel, or in some cases omitted. Likewise, the order of processing is not necessarily required to achieve the features and advantages of the example embodiments described herein, but is provided for ease of illustration and description. One or more of the illustrated actions, operations, and/or functions may be repeatedly performed depending on the particular strategy being used. Further, the described actions, operations, and/or functions may graphically represent code to be programmed into non-transitory memory of the computer readable storage medium in the control system, where the described actions are carried out by executing the instructions in a system including the various hardware components in combination with the hub controller and/or the fire device controller.

(74) It will be appreciated that the configurations and routines disclosed herein are exemplary in nature, and that these specific embodiments are not to be considered in a limiting sense, because numerous variations are possible. Moreover, unless explicitly stated to the contrary, the terms “first,” “second,” “third,” and the like are not intended to denote any order, position, quantity, or importance, but rather are used merely as labels to distinguish one element from another. The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various systems and configurations, and other features, functions, and/or properties disclosed herein.

(75) As used herein, the term “approximately” is construed to mean plus or minus five percent of the range unless otherwise specified.

(76) The following claims particularly point out certain combinations and sub-combinations regarded as novel and non-obvious. These claims may refer to “an” element or “a first” element or the equivalent thereof. Such claims should be understood to include incorporation of one or more such elements, neither requiring nor excluding two or more such elements. Other combinations and sub-combinations of the disclosed features, functions, elements, and/or properties may be claimed through amendment of the present claims or through presentation of new claims in this or a related application. Such claims, whether broader, narrower, equal, or different in scope to the original claims, also are regarded as included within the subject matter of the present disclosure.

Claims

1. A fire display system, comprising: a fuel source comprising gaseous fuel; a common fuel line coupled to the fuel source via a first pressure regulator providing the gaseous fuel to the common fuel line at a first pressure; and a first fire display device fluidically coupled to the common fuel line, wherein the first fire display device is coupled to the common fuel line via a second pressure regulator arranged downstream of the first pressure regulator and providing the gaseous fuel to the first fire display device at a second pressure that is lower than the first pressure, wherein a second fire display device is fluidically coupled to the common fuel line via a third pressure regulator arranged downstream of the first pressure regulator and providing the gaseous fuel to the second fire display device at a third pressure that is lower than the first pressure.
2. The fire display system of claim 1, wherein the third pressure is equal to the second pressure.
3. The fire display system of claim 1, wherein the third pressure is a different pressure, lower than the second pressure.
4. A fire display system, comprising: a fuel source comprising gaseous fuel; a common fuel line

coupled to the fuel source via a first pressure regulator providing the gaseous fuel to the common fuel line at a first pressure; a first fire display device fluidically coupled to the common fuel line, wherein the first fire display device is coupled to the common fuel line via a second pressure regulator arranged downstream of the first pressure regulator and providing the gaseous fuel to the first fire display device at a second pressure that is lower than the first pressure; and a bleed valve coupled to the common fuel line that is configured to vent air out of the common fuel line.

5. The fire display system of claim 4, wherein the bleed valve is positioned at an opposite end of the common fuel line as the fuel source.

6. A fire display system, comprising: a fuel source comprising gaseous fuel; a common fuel line coupled to the fuel source via a first pressure regulator, wherein the first pressure regulator is configured to provide the gaseous fuel to the common fuel line at a first pressure; and a plurality of fire display devices each fluidically coupled to the common fuel line via respective pressure regulators, the respective pressure regulators configured to flow the gaseous fuel from the common fuel line to the plurality of fire display devices at pressures less than the first pressure, wherein the respective pressure regulators include a second pressure regulator, and wherein the second pressure regulator is positioned between the common fuel line and a first fire display device of the plurality of fire display devices.

7. The fire display system of claim 6, further comprising a bleed valve coupled to the common fuel line, wherein the bleed valve is downstream of the plurality of fire display devices.

8. The fire display system of claim 7, wherein the bleed valve is configured to release air out of the common fuel line.

9. The fire display system of claim 6, wherein the plurality of fire display devices are each further fluidically coupled to the common fuel line via respective check valves.

10. The fire display system of claim 6, wherein a fuel line extends between the second pressure regulator and the first fire display device.

11. The fire display system of claim 6, wherein each of the plurality of fire display devices comprises: a burner; an electrically actuatable gas valve configured to adjust an amount of gaseous fuel flowed to the burner for ignition; a wireless receiver; and a controller, wherein the controller comprises instructions stored in non-transitory memory of the controller that are executable to: adjust a position of the electrically actuatable gas valve responsive to a request received via the wireless receiver.

12. The fire display system of claim 11, wherein the request received via the wireless receiver is from a hub of the fire display system.

13. A method for operating a multi flame display, comprising: providing gaseous fuel to a common fuel line at a first pressure via a first common fuel line pressure regulator; and flowing the gaseous fuel to a plurality of flame display devices, wherein the plurality of flame display devices are fluidically coupled to the common fuel line downstream via respective pressure regulators, wherein the respective pressure regulators are positioned downstream the first common fuel line pressure regulator and provide the gaseous fuel to the plurality of flame display devices from the common fuel line to the plurality of flame display devices at lower pressures than the first pressure, the method further comprising igniting the gaseous fuel flowed to the plurality of flame display devices responsive to commands received at the plurality of flame display devices from a hub that are coordinated to an audio input.

14. The method of claim 13, further comprising, after providing the gaseous fuel to the common fuel line at the first pressure and prior to flowing the gaseous fuel to the plurality of flame display devices, venting air trapped in the common fuel line via a bleed valve.

15. The method of claim 14, wherein venting the air trapped in the common fuel line includes opening the bleed valve to release the air out of the common fuel line.

16. The method of claim 13, wherein the commands are received wirelessly from the hub, and wherein igniting the gaseous fuel flowed to the plurality of flame display devices includes adjusting

a position of an electric valve included in each of the plurality of flame display devices.

17. The method of claim 16, wherein the commands received at a first flame display device differs from the commands received at a second flame display device of the plurality of flame display devices.
